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Multimodal imaging and AI in brain tumor surgery: current tools and emerging integration



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Brain tumors remain among the most challenging malignancies, necessitating advances in diagnostic accuracy and treatment precision. Integrated multimodal surgical platforms combining structural, functional, and molecular data are transforming neuro-oncology by improving surgical planning, intraoperative guidance, and patient monitoring. This perspective reviews current and emerging imaging modalities including optical techniques and AI-integrated platforms, while discussing their integration within unified clinical workflows that may advance personalized brain tumor care and surgical outcomes.

Brain tumors represent a major global health burden, characterized by high morbidity and mortality despite advances in diagnosis and treatment. According to recent GLOBOCAN estimates, tumors of the brain and central nervous system (CNS) account for approximately 322,000 new cases and 248,000 deaths worldwide each year, highlighting their disproportionately high lethality¹. In adults, primary brain tumors include gliomas, meningiomas, and other less common pathologies, all of which demonstrate substantial heterogeneity in phenotypic tumor behavior and associated prognosis, as well as the potential for highly aggressive malignant forms, many of which demonstrate a five-year relative survival of less than 35% in population-based studies². In parallel, brain metastases are cumulatively the most prevalent intracranial tumor in adults, affecting an estimated 15–40% of patients with advanced cancers, in particular those with primary lung, breast, or melanoma pathologies³. Brain metastases are increasing in incidence, attributable in part to advances in systemic therapies, improved surveillance, and broader access to advanced imaging⁴. In this context, accurate tumor detection, characterization, treatment planning, and response assessment remain critical challenges for contemporary neuro-oncology. Multimodal surgical platforms integrating structural, functional, and molecular techniques have emerged as essential tools to tackle these challenges by synergizing data on tumor anatomy, metabolism, vascularity, and the local CNS microenvironment, empowering precision diagnostics and individualized therapeutic strategies.

The technologies discussed in this perspective span a spectrum of clinical maturity, from well-established modalities such as MRI, PET, 5-aminolevulinic acid, and intraoperative ultrasound, which are supported by robust evidence bases, to emerging translational technologies including

laser speckle contrast imaging (LSCI), hyperspectral imaging, and AI-driven multimodal integration, which carry promising early clinical data but limited large-scale validation, to approaches that remain largely investigational, including AI-integrated platforms capable of simultaneous intraoperative fusion of all imaging modalities, and MR-guided focused ultrasound for targeted disruption of the blood-brain barrier to enhance chemotherapeutic delivery. This stratification is reflected in Table 1 and provides important context for interpreting the evidence discussed in each section. It should also be noted that the evidence base supporting many of these modalities is predominantly derived from glioma surgery; applicability to meningiomas, brain metastases, and other tumor types varies by modality and is explicitly acknowledged where relevant.

In this perspective, we review the current applications of integrated multimodal surgical platforms for brain tumor diagnosis and treatment. We discuss the established clinical imaging modalities of magnetic resonance imaging (MRI) and positron emission tomography (PET), as well as advances in the intraoperative applications of MRI and ultrasound. We discuss the emerging optical imaging techniques 5-aminolevulinic acid, indocyanine green angiography, LSCI, and hyperspectral imaging. Finally, we examine the growing role of artificial intelligence (AI) and machine learning (ML) in multimodal data integration and predictive modeling. The components of an integrated surgical intelligence framework, spanning structural, functional, molecular, and optical data streams, a computational integration layer enabling AI-driven fusion and real-time analysis, intraoperative decision support outputs, and downstream implications for post-operative oncologic care, are discussed collectively with the goal of illustrating how these tools interact within a unified clinical workflow.

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Table 1 | Key advantages, limitations, and clinical context of imaging modalities for neuro-oncology treatment

Modality	Clinical maturity	Primary use case	Primary tumor evidence	Workflow burden	Key advantages	Key limitations	Complementary Role
MRI	Established	Preop planning; postop monitoring	All tumor types	Low (outpatient)	High spatial resolution; excellent soft-tissue contrast; clinical backbone	Cannot fully characterize metabolism, hypoxia, or immune activity; nonspecific for pseudoprogression	Reference standard for all other modalities
PET	Established	Preop planning; treatment response	Glioma (amino acid); meningioma (somatostatin)	Low–Medium	Adds metabolic/molecular information; identifies infiltrative tumor beyond MRI	Requires MRI for anatomical context; radioisotope exposure; high FDG background signal	Complements MRI for grading and recurrence assessment
iMRI	Established	Intraoperative guidance	Glioma	High	Gold standard for residual tumor detection; compensates for brain shift	50–80+ min workflow disruption; high cost; infection risk	Benchmark for intraoperative tools
iUS	Established	Intraoperative guidance	Glioma	Low	Real-time; portable; affordable; no ionizing radiation	Operator-dependent; noisier images	Cost-effective complement to iMRI
5-ALA	Established	Intraoperative guidance	High-grade glioma	Low	Improves extent of resection; low cost; ease of use	Primarily validated in high-grade glioma; requires specialized filters	Synergistic with iMRI and iUS
ICG-VA	Established	Intraoperative guidance	Glioma; cerebrovascular	Low	Real-time vascular flow visualization; prevents iatrogenic vascular injury	Requires exogenous contrast agent; primarily qualitative	Complements LSCI for perfusion assessment
LSCI	Emerging translational	Intraoperative guidance	Glioma; cerebrovascular	Low–Medium	Continuous label-free perfusion monitoring; no workflow interruption	Restricted commercial availability; requires quantitative modeling; limited prospective validation	Complements ICG-VA; candidate for AI integration
HSI	Emerging translational	Intraoperative guidance	Glioblastoma; metastases	Medium	High-precision tissue differentiation; non-invasive; label-free	Limited acquisition speed; high inter-patient variability; limited multicenter validation	Potential integration with fluorescence-guided surgery
AI/ML	Emerging translational – Investigational	All phases	Predominantly glioma	Variable	Enables multimodal data fusion; radiogenomics; automated segmentation; outcome prediction	Generalizability; interpretability; regulatory barriers; workflow integration	Cross-cutting integrative layer
MRgFUS	Investigational	Adjuvant therapy delivery	High-grade glioma (early data)	High	Targeted BBB disruption for drug delivery	Limited clinical data; infrastructure requirements	Future integration with intraoperative platforms

Brain tumor biology

Brain tumors exhibit marked spatial, temporal, and biological heterogeneity, which fundamentally limits the accuracy of diagnosis, prognosis, and treatment response assessment^{5,6}. This heterogeneity is evident both across tumor types, such as glioblastoma, meningioma, and brain metastases, and within individual lesions, where distinct cellular subclones, vascular niches, metabolic states, and immune microenvironments coexist and dynamically evolve. In glioblastoma, regional differences in cellularity, hypoxia, angiogenesis, and molecular alterations are well-recognized contributors to therapeutic resistance and inevitable recurrence⁶. Similarly, meningiomas may demonstrate heterogeneous growth patterns, vascularity, and biological aggressiveness that are not reliably predicted by histopathology alone^{7,8}, while brain metastases reflect diverse primary tumor biology and complex interactions with the brain microenvironment⁹. Although the current WHO classification emphasizes molecular features as central diagnostic criteria, tissue sampling remains spatially limited and temporally static, highlighting the need for complementary, non-invasive approaches capable of capturing tumor-wide and longitudinal biological complexity^{10,11}.

Magnetic resonance imaging (MRI) and positron emission tomography (PET)

MRI remains the clinical backbone of brain tumor evaluation owing to its high spatial resolution and highly granular soft-tissue characterization¹². Conventional MRI sequences are indispensable for anatomical delineation, surgical planning, and longitudinal follow-up; however, single-modality imaging approaches are insufficient to represent the multidimensional nature of tumor biology. Structural MRI fails to fully characterize key

processes such as cellular proliferation, metabolism, angiogenesis, blood–brain barrier dysfunction, hypoxia, and immune activity¹³. Even advanced MRI techniques, such as diffusion-weighted imaging (DWI), perfusion imaging, susceptibility-based methods, and MR spectroscopy, are relatively nonspecific and often inadequate for clinically challenging and commonplace scenarios such as the accurate differentiation of pseudo-progression, radiation necrosis, or post-treatment inflammation from true disease recurrence/progression^{14,15}.

Integrated multimodal surgical platforms provide a transformative framework by enabling integrated assessment of tumor structure, function, metabolism, and molecular activity across spatial and temporal scales. The combination of anatomical MRI with functional and molecular imaging modalities, like PET using glucose metabolism tracers, demonstrates added value for tumor grading, delineation of biologically active tumor regions, and differentiation of treatment-related changes from true progression¹⁶. A key limitation of classic fludeoxyglucose-18 (FDG) PET in neuro-oncology is the inherently high physiologic glucose uptake of normal brain tissue, which creates substantial background signal that obscures tumor detection and limits diagnostic specificity. This has driven a shift toward amino acid tracers which exploit the upregulated amino acid transport characteristic of tumor cells while producing minimal uptake in healthy brain parenchyma, resulting in significantly improved tumor-to-background contrast^{16,17}. In gliomas, PET using amino acid tracers frequently identifies metabolically active infiltrative tumor beyond contrast-enhancing regions on MRI, while in brain metastases it improves target definition for stereotactic radiosurgery¹⁸. In meningiomas, somatostatin receptor-based PET has shown high sensitivity for lesion detection and treatment planning,

particularly in complex skull-base disease, though the evidence base in this tumor type is more limited compared to gliomas^{16,19}.

Recent advances in imaging hardware, including hybrid PET/MRI scanners, ultra-high-field MRI, and accelerated acquisition techniques, have converged with rapid progress in computational analysis to create a critical inflection point for clinical translation²⁰. AI-driven image fusion, radiomics, and deep learning-based multimodal integration allow the extraction of high-dimensional imaging features that capture complex spatial patterns and cross-modal relationships beyond human visual interpretation^{21,22}. These approaches enable the development of non-invasive imaging biomarkers linked to tumor genotype, phenotype, and clinical outcomes, advancing the paradigm of radiomics and imaging-based tumor profiling.

Intraoperative MRI (iMRI) and intraoperative ultrasound (iUS)

Intraoperative imaging modalities have become central to maximizing safe resection of brain tumors, particularly diffuse gliomas, by providing real-time guidance that complements preoperative planning. Intraoperative MRI (iMRI) allows high-resolution anatomical updates during surgery, enabling detection of residual tumor and facilitating real-time adjustment of resection strategies²³. iMRI is widely regarded as a core strategy for optimizing extent of resection in gliomas, offering precise detection of residual tumor prior to the conclusion of surgery, as well as updated stereotaxis that compensates directly for brain shift after initial resection²⁴. High-field (1.5, 3T) systems provide excellent resolution for distinguishing tumor from healthy tissue and eloquent areas, often leading to higher gross total resection rates and improved progression-free survival in high-grade and low-grade gliomas. However, iMRI has notable disadvantages: it causes significant workflow disruption (typically 50–80+ minutes per scan due to patient transport, field preparation, and scanning), increases operative time and anesthesia exposure, raises infection risk from prolonged interruptions, and incurs high acquisition, maintenance, and infrastructure costs, limiting adoption outside high-volume, high-resource centers^{25,26}.

In contrast, iUS offers real-time, repeatable imaging that integrates seamlessly into the surgical workflow with minimal disruption, providing immediate feedback on tumor boundaries, resection cavities, and brain shift without requiring patient repositioning. It is portable, affordable, and safe, with no ionizing radiation, making it more accessible and cost-effective than iMRI²⁷. While iUS images are noisier and more operator-dependent, navigated 3D iUS fused with preoperative MRI compensates effectively for brain shift, achieving target registration errors below 2–3 mm and supporting dynamic updates^{28,29}. Intraoperative contrast-enhanced ultrasound using microbubbles has emerged as a valuable tool for glioma surgery, providing real-time visualization of tumor perfusion and vascularized margins beyond conventional US imaging³⁰. Microbubble contrast agents remain intravascular, enhancing tissue perfusion signals and allowing identification of residual tumor intraoperatively. Clinical studies suggest that iUS may improve extent of resection and help differentiate tumor from normal brain, particularly in hypervascular regions, potentially impacting progression-free survival^{31,32}.

Neuronavigation systems

Neuronavigation systems integrate preoperative imaging with real-time updates of surgical anatomy, improving localization and minimizing damage to eloquent brain regions. The accuracy of neuronavigation systems is increasingly enhanced by AI-based algorithms that compensate for intraoperative brain shift³³, enabling more accurate and dynamic alignment between preoperative imaging and the evolving intraoperative surgical field throughout the course of resection. Awake craniotomy with direct electrical stimulation (DES) facilitates detection of eloquent brain areas, while asleep cortical mapping or subcortical stimulation allow for intraoperative detection of functionally critical regions. Intraoperative imaging modalities, individually and in combination, create a synergistic framework that enhances maximal safe resection, integrates biological and functional

information, and sets the stage for AI-assisted, multimodal, real-time surgical decision-making in glioma surgery^{34,35}.

The integration of iUS with neuronavigation systems, preoperative MRI, and AI further enhances precision during brain tumor resection. Commercial platforms now incorporate fusion algorithms to align iUS volumes with MRI. AI-driven tools automate tumor segmentation, reduce speckle artifacts, enable deformable registration, and estimate brain shift in real time, thereby reducing operator dependency and improving consistency. Multicenter validation studies using the BraTioUS dataset demonstrate that AI-enhanced iUS can accurately delineate gliomas, with high segmentation accuracy on internal testing and promising generalizability across independent iUS systems³⁶. While segmentation strategies would benefit from further improvements in accuracy and intraoperative implementation, evidence supports iUS-MRI-AI integration as a cost-effective and workflow-friendly approach to intraoperative guidance, capable of achieving extent-of-resection outcomes comparable to iMRI in appropriately selected cases³⁶.

Emerging optical imaging modalities

5-Aminolevulinic acid (5-ALA)

5-Aminolevulinic acid (5-ALA) is one of the main fluorophores used during fluorescence-guided surgery (FGS) for intra-axial brain tumor resection. Multiple studies have demonstrated that 5-ALA guided surgery improves the extent of resection compared with conventional white-light microsurgery and is associated with improved progression-free survival in patients with high-grade glioma^{37,38}. Due to its relative ease of use and low cost, 5-ALA has emerged as a cornerstone of fluorescence-guided surgery, and its integration with complementary intraoperative modalities such as iMRI and iUS within multimodal platforms is an active area of clinical and technical development. The finding that 5-ALA guidance achieves results comparable to iMRI³⁹ highlights its value as a practical, cost-effective core technology that, when integrated with iMRI and iUS, can provide synergistic benefits beyond what either modality achieves independently.

Indocyanine green video angiography (ICG-VA) and FLOW 800/ GLOW800

Indocyanine green video angiography (ICG-VA) is an intraoperative fluorescence imaging technique to assess vascular patency and blood flow in real time, particularly during neurovascular procedures⁴⁰. Since its introduction in 2003, ICG-VA has become ubiquitous among cerebrovascular operations such as aneurysm clipping or AVM resection, given its outstanding capacity for real-time visualization of arterial and venous structures and flow dynamics in the surgical field⁴¹. Leading neurosurgical microscope manufacturers Zeiss and Leica provide microscope-integrated software (e.g., FLOW 800 and GLOW800) to perform quantitative analyses of cerebral blood flow based on ICG-VA. Recent studies evaluating the use of FLOW 800 or GLOW800 during brain tumor resection surgeries demonstrated additional utility for real-time assessment of brain perfusion, as well as avoidance of iatrogenic vascular injury^{42,43}. Indocyanine green has also been successfully used to localize tumor margins following a high-dose administration 24 hours before surgery⁴⁴. Growing evidence supports the expanded use of ICG-VA and associated quantitative analyses to better characterize brain perfusion and detect hypoperfusion or hyperperfusion phenomena at the conclusion of tumor resection. Integration of ICG-VA with modalities offering more continuous quantitative perfusion assessment, such as LSCI, represents a complementary approach that may amplify the utility of both tools^{45,46}.

Laser speckle contrast imaging (LSCI)

Laser speckle contrast imaging (LSCI) is emerging as a promising tool to evaluate cerebral blood flow and tissue perfusion during neurosurgical procedures⁴⁷. Unlike ICG-VA, LSCI does not require an exogenous contrast agent and thus enables continuous, real-time intraoperative monitoring of blood flow dynamics⁴⁸. Clinical studies have demonstrated that LSCI provides blood flow assessment capabilities comparable to ICG-VA with the

additional advantage of repeated, non-invasive measurements without interruption of the surgical workflow^{45,46}. Another advantage is that LSCI can assess tissue perfusion, enabling detection of cortical hypoperfusion and hyperperfusion phenomena that may occur during tumor resection or vessel manipulation. LSCI has demonstrated utility for continuous, label-free perfusion monitoring during neurosurgical procedures, with ongoing work focused on integration as an on-board package with commercially available surgical microscopes and development of quantitative modeling approaches aimed at estimating absolute blood flow. However, several important limitations currently constrain its broader adoption. Commercial availability remains restricted, with no widely deployed standalone surgical system yet cleared for this indication. Accurate estimation of absolute cerebral blood flow requires quantitative biophysical modeling, adding computational complexity to intraoperative workflows. Prospective clinical validation studies in large, heterogeneous patient populations remain limited, and multicenter data are needed to establish generalizability across different surgical environments, tumor types, and patient characteristics. These barriers notwithstanding, LSCI represents a promising approach for continuous perfusion monitoring during brain tumor surgery, particularly when integrated with complementary modalities such as ICG-VA.

Hyperspectral imaging (HSI)

Hyperspectral imaging (HSI) is an emerging optical technology that captures detailed spectral data across hundreds of wavelengths, enabling precise tissue differentiation during neurosurgical operations. In human brain tumor surgeries, HSI addresses limitations of conventional tools like preoperative MRI, where stereotactic accuracy is highly susceptible to error from brain deformation that occurs because of bone removal, tumor resection, or tissue edema, among other factors. By analyzing reflectance or fluorescence spectra, HSI identifies tumor margins, distinguishes infiltrative gliomas from healthy tissue, and maps perfusion or oxygenation in real-time. Key systems include snapshot cameras mounted on surgical microscopes, operating in visible-near-infrared ranges for rapid acquisition. Computational methods such as machine learning classifiers have demonstrated pixel-wise tumor segmentation accuracy exceeding 99% agreement with histopathology in certain controlled experimental settings; however, this figure derives from specific acquisition conditions and selected tumor types, and performance varies substantially across different tumor histologies, imaging systems, and patient populations, with limited prospective validation in heterogeneous clinical cohorts to date⁴⁹. Clinical studies focusing primarily on glioblastoma and metastases resections have demonstrated promising avenues for HSI deployment as a potentially high-impact strategy for optimizing both extent-of-resection and preservation-of-function. This is accomplished by detecting low levels of protoporphyrin IX (PpIX) fluorescence, a naturally occurring compound that accumulates in tumor cells and fluoresces when excited with violet-blue light⁵⁰. A foundational technical validation study by Gebhart et al.⁵¹ first applied HSI for brain tumor demarcation using tunable filters, demonstrating that HSI could discriminate between brain tissues⁵¹. Advantages of HSI include its non-invasiveness and label-free operation (eliminating the need for injected dyes or contrast agents), although challenges such as acquisition speed and inter-patient tumor variability continue to exist.

How have AI and ML shaped brain tumor imaging techniques and insights?

Machine learning algorithms and artificial neural networks offer new insights into biomarker and drug discovery, radiogenomics, rapid pathologic evaluation, and image-based assessments of treatment response⁵². Models aim to provide a deeper understanding and improved management of neuro-oncologic disease, and imaging is a domain of particularly rapid integration. Prior to diagnosis, AI-based tools are highly accurate in classifying brain regions extensively involved with infiltrating tumor from adjacent, healthy, non-malignant tissue⁵³ and may predict tumor grade preoperatively^{54,55}. The growing field of radiogenomics has the potential to reveal key molecular signatures such as O6-methylguanine-DNA

methyltransferase (MGMT) promoter methylation status, a marker with predictive and prognostic significance in high-grade glioma⁵⁶, 1p/19q-codeletion and isocitrate dehydrogenase (IDH) mutation status, characteristics of oligodendroglioma^{57,58}, H3K27M-mutated diffuse midline glioma⁵⁹, and other genetic aberrations that are characteristic of glioblastoma such as TERT promoter⁶⁰, PTEN⁶¹, and TP53⁶¹ mutations. These results are particularly valuable for surgically inaccessible or technically challenging lesions, may guide extent-of-resection, and can complement circulating tumor DNA testing. Complementing these AI-driven imaging approaches, emerging intraoperative tools such as stimulated Raman histology-based microscopy can identify tumor invasion at a histological level beyond what MRI demonstrates⁶², though widespread clinical adoption remains limited by its current availability at only a small number of highly specialized centers.

Once a brain tumor diagnosis has been established and protocol-based treatment initiated, a critical aspect of care is monitoring response to treatment. Automated and semiautomated tumor segmentation tools^{63–65} substantially reduce per-MRI review time⁵⁴ and quantify brain involvement by primary and metastatic neoplasms in two-dimensional and volumetric outputs. Building upon validated segmentation algorithms, automated pipelines evaluate longitudinal response to therapy, which can inform standard of care and clinical trial assessments^{66,67}. Meanwhile, a critical area of ongoing research is radiographic determination of radiation-induced pseudoprogression versus true disease progression, a common and impactful clinical challenge for many glioblastoma patients⁶⁸.

Prognostication remains an elusive component of brain tumor care in clinical practice. Predicting response of brain metastases to stereotactic radiosurgery represented early efforts⁶⁹, and there is ongoing interest in survival prediction across clinically and molecularly variable tumors⁷⁰. Emerging connectomics research with diffusion MRI has demonstrated that glioblastoma disrupts white matter connectivity far beyond the visible tumor margins. The extent of structural disruptions is associated with shorter survival and represents a potential target for precise treatment, reinforcing the need for integrated multimodal platforms capable of capturing tumor-related changes beyond conventional imaging⁷¹. These insights, supplementing molecular features of the tumor, may help neuro-oncologists understand disease processes and communicate more effectively with patients.

Despite these advances, significant barriers limit the clinical translation of AI tools in neuro-oncology. Most published models have been developed and validated on single-institution datasets, raising important concerns about generalizability across centers, imaging platforms, and patient populations. Domain shift, the degradation in model performance when applied to data from different acquisition protocols or scanner types, remains a pervasive and underappreciated challenge. Dataset bias, including underrepresentation of diverse patient populations and tumor subtypes, further constrains the applicability of published results. Model interpretability is a central concern for clinical adoption: black-box algorithms operating on high-dimensional imaging data are difficult to audit, explain to clinicians, or defend in medico-legal contexts. Reproducibility of radiomics and deep learning results across independent cohorts has been inconsistent. Regulatory approval pathways for AI-based diagnostic and prognostic tools remain evolving and vary substantially across jurisdictions, creating uncertainty for developers and health systems alike. Finally, seamless integration of AI tools into intraoperative workflows, where latency, interface design, and cognitive load are critical, has received comparatively little attention and represents a major practical barrier to real-world deployment. Addressing these challenges will be essential for AI to fulfill its role as the integrative layer of unified multimodal surgical platforms.

The future of integrated multimodal surgical platforms for brain tumors

The principle of complementarity that underlies integrated multimodal platforms is best illustrated through the perioperative management of high-grade glioma. Preoperatively, structural MRI defines tumor anatomy and

informs surgical planning, while amino acid PET delineates metabolically active infiltrative regions beyond the contrast-enhancing core, together establishing resection boundaries that neither modality could define alone. Intraoperatively, iUS provides real-time anatomical updates that compensate for progressive brain shift, while 5-ALA fluorescence identifies residual tumor at resection margins that white-light microscopy cannot reliably detect. Concurrently, LSCI enables continuous perfusion monitoring to identify hypoperfusion or inadvertent vascular compromise, a capability that ICG-VA, as a single-timepoint contrast injection, cannot provide. AI-driven segmentation algorithms operating on fused intraoperative data streams synthesize these spatial and functional inputs to generate decision support at the point of care. Postoperatively, advanced MRI sequences assess extent of resection and monitor for early recurrence, while AI-based longitudinal analysis of serial imaging supports both standard-of-care response assessment and clinical trial endpoints. In each phase of this workflow, each modality directly addresses a specific limitation of the preceding one—a relationship of functional complementarity, rather than redundancy, that constitutes the defining rationale for integrated multimodal platforms.

Building on the complementary roles described above, the trajectory of multimodal surgical platforms points toward systems in which structural, functional, molecular, and optical data are continuously fused and interpreted in real time at the point of care. AI will serve as the integrative layer enabling this synthesis, automating segmentation, compensating for brain shift, detecting residual tumor, and flagging perfusion compromise simultaneously. Beyond intraoperative guidance, these platforms are expected to generate longitudinal data that inform postoperative therapy selection and treatment response assessment. Achieving this will require progress across hardware miniaturization, computational latency reduction, multicenter dataset development, and regulatory frameworks adapted to intraoperative AI, challenges that are active areas of development across the field.

The potential of these platforms extends beyond intraoperative guidance to adjuvant therapy delivery. MR-guided focused ultrasound (MRgFUS) with microbubbles exemplifies this convergence, enabling transient, targeted disruption of the blood-brain barrier to enhance penetration of chemotherapeutic agents such as temozolomide into infiltrative tumor regions. A landmark multicenter phase 1/2 trial demonstrated the safety and feasibility of this approach in high-grade glioma, with survival outcomes substantially outperforming historical controls⁷², and neuronavigation-guided FUS has since shown safety in pediatric diffuse midline gliomas⁷³. These early results suggest that integrated surgical platforms may ultimately support not only the resection itself but the perioperative oncologic strategy as a whole, connecting preoperative planning, intraoperative guidance, and adjuvant therapy optimization within a single, coherent framework of care.

Data availability

No datasets were generated or analyzed during the current study.

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Author contributions

D.P.C. and D.R.M.: concept design. D.P.C., T.A.N., and D.R.M.: writing first draft. All authors (D.P.C., T.A.N., C.S.G., J.B., D.R.M.) provided critical review and commentary of revised drafts and read and approved the final manuscript.

Competing interests

Co-author Thomas A. Nelson serves on the advisory boards for Grai Med, Inc, and Wellytics, Inc. Co-author David R. Miller declares intellectual property rights in the area of laser speckle contrast imaging, including a

provisional patent application (Application No. 63/921,815, filed November 20, 2025; title: *Method for Predicting Complications Related to Blood Flow or Tissue Perfusion During Surgery*; applicant: University of Oklahoma; inventor: David R. Miller; status: pending), and holds stock in, serves on the board of directors, and consults for Dynamic Light. The other authors do not have a competing interest.

Additional information

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